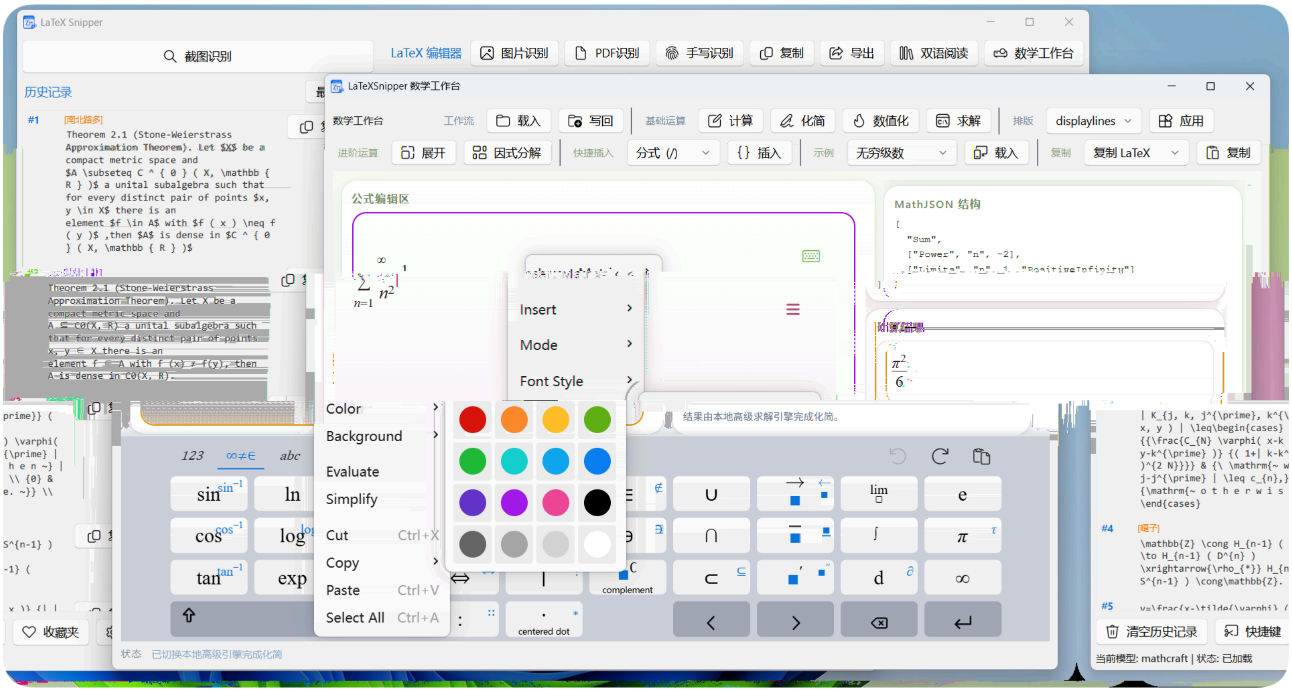
LaTeXSnipper



- **LaTeXSnipper**

- **MathCraft**

· LaTeXSnipper

⚠️

`python3 -m venv` `python3-venv`

! 80%

→ → **/** →

Windows `%USERPROFILE%\.latexsnipper\logs\` `%LOCALAPPDATA%\LaTeXSnipper\logs\`

Linux `~/.latexsnipper/logs/`

macOS `~/.latexsnipper/logs/`

`crash-native.log`

✔️

`Ctrl+F`

PDF

/

/

+

PDF

PDF

Poppler **Fitz**

Argos Translate **Azure Translator** **Google Cloud Translation**

DeepL API Free

GPU MATHCRAFT_FORCE_ORT_CPU=1

GPU

/

invalid protobuf failed to load model sha256 mismatch

%APPDATA%\MathCraft\models\<model_id>\

~/.mathcraft/models/<model_id>/

python -m mathcraft_ocr models check

ONNX

MATHCRAFT_FORCE_ORT_CPU=1

✗

_validate_config() model_name base_url

model_name

→

A

B

C ollama run <model_name>

Ollama Base URL

http://localhost:11434/v1 /v1

https:// http://

ollama serve

✔

http://127.0.0.1:11434/api/tags

/

ollama list

/v1/models

API API Key

⚠

MinerU

`/health` `/file_parse`

`http://127.0.0.1:8000/health`

OpenAI-compatible Ollama

`/v1/models`

`/api/tags` `/v1/models`

!

Ollama

OpenAI-compatible

MinerU

Python

Windows

Linux/macOS

/

```
创建虚拟环境

激活

激活

安装依赖
```

PyInstaller

`git clone`

`requirements.txt` `requirements-build.txt`

Linux/macOS

`tools/deps/python311`

```
```

> **!**
>
> `/usr/lib/latexsnipper` `.app`
>
> `~/.latexsnipper/deps/python311`
>
> `pyvenv.cfg`

`.deb` `python3` `python3-venv` `.deb`

`python3`

```
```

或使用 官方 安装包

pip pywin32 / PyQt6

pip install -r requirements.txt

ensurepip pip setuptools wheel

python-3.11.0-amd64.exe

pywin32

PyQt6

pywin32

Program Files /

C:\Program Files\

%APPDATA% ~/.latexsnipper/

%LOCALAPPDATA%

/

%APPDATA%

MATHCRAFT_HOME C:\mathcraft_data

Windows

`0.0.0.0` `127.0.0.1`

`api.github.com`

`%USERPROFILE%/.latexsnipper/logs/`

/

GitHub Releases

PDF

PDF

/

Linux / macOS

Pandoc

/

/

`ctexart`

`amsmath` `amssymb` `mathtools` `geometry`

*

`2x` `2*x`

/

Pandoc

`.docx` `.epub`

`ctex`

PDF

PDF

Linux / macOS

---

pynput

Ctrl+字母 Ctrl+Shift+字母
Ctrl+F
grim maim gnome-screenshot
screencapture

/

~/.latexsnipper/deps/python311

.deb .dmg .app.zip

> **!**
>
> Ctrl+字母 Ctrl+Shift+字母

Linux: Wayland

grim slurp gnome-screenshot

> **! Linux**
>
> grim maim gnome-screenshot

Linux: Wayland

EGL not available No physical devices GLOzone not found Failed to get system egl display

/dev/dri/renderD*

```
强制启用 图形兜底

禁用 图形兜底，尝试原生 路径
```

macOS:

xattr -cr /Applications/LaTeXSnipper.app

macOS:

macOS:

macOS: pip pip

pip pip setuptools wheel

~/.latexsnipper/deps/python311 venv ensurepip pip

brew install python

配置文件　　日志:

依赖环境:

模型缓存（　　　　　　　　）:

Python

`where python` `which python`

/

Bug

✗
“ ”“ ”“ ” issue

1.
%USERPROFILE%\.latexsnipper\logs\ %LOCALAPPDATA%\LaTeXSnipper\logs\ **logs**

2.
crash-native.log

3.

4.

5.

6.

GitHub Issues
GitHub Discussions

README Quick Start

Windows `python -m venv .venv` `pip install -r requirements.txt` `python src/main.py`
Linux `python3 -m venv .venv` `pip install -r requirements-linux.txt` `python src/main.py`
macOS `python3 -m venv .venv` `pip install -r requirements-macos.txt` `python src/main.py`
`requirements-linux.txt` `requirements-macos.txt` `-r requirements.txt`
`pynput`

`python311/` `tools/deps/python311/`

`python311/`

`tools/deps/python311/` `requirements*.txt`
`ruff` `pytest` `pyright`
`tools/deps/python311` `~/.latexsnipper/`
`deps/python311`

Windows

!
`python311/`
`tools/deps/python311/`
`~/.latexsnipper/deps/python311`

`docs/developer_code_standards.md`

⚠
`ruff` `pytest` `pyright` `requirements*.txt`

pyrightconfig.json

tools/deps/python311 python311/

tools/deps/python311 python311

· MathCraft

MathCraft OCR

ONNX

```
检查模型缓存状态

手动下载模型

识别图片中的公式
```

LaTeXSnipper

mathcraft-ocr 0.2.0 v1.0.0 **4**

ONNX

- mathcraft-formula-det
- mathcraft-formula-rec
- mathcraft-text-det
- mathcraft-text-rec

> **!**

Profiles

```
formula
text
```

```
mixed
```

Abstract Algebra (p.18)

PROOF. (1) If $A \cup B = \emptyset$, then the theorem follows immediately since both A and B are the empty set. Otherwise, we must show that $(A \cup B)' \subset A' \cap B'$ and $(A \cup B)' \supset A' \cap B'$. Let $x \in (A \cup B)'$. Then $x \notin A \cup B$. So x is neither in A nor in B, by the definition of the union of sets. By the definition of the complement, $x \in A'$ and $x \in B'$. Therefore, $x \in A' \cap B'$ and we have $(A \cup B)' \subset A' \cap B'$.

To show the reverse inclusion, suppose that $x \in A' \cap B'$. Then $x \in A'$ and $x \in B'$, and so $x \notin A$ and $x \notin B$. Thus $x \notin A \cup B$ and so $x \in (A \cup B)'$. Hence, $(A \cup B)' \supset A' \cap B'$ and so $(A \cup B)' = A' \cap B'$.

The proof of (2) is left as an exercise.

Example 1.4 Other relations between sets often hold true. For example,

$$(A \setminus B) \cap (B \setminus A) = \emptyset.$$

To see that this is true, observe that

$$\begin{aligned}(A \setminus B) \cap (B \setminus A) &= (A \cap B') \cap (B \cap A') \\ &= A \cap A' \cap B \cap B' \\ &= \emptyset.\end{aligned}$$

Cartesian Products and Mappings

Given sets A and B, we can define a new set $A \times B$, called the ***Cartesian product*** of A and B, as a set of ordered pairs. That is,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Example 1.5 If $A = \{x, y\}$, $B = \{1, 2, 3\}$, and $C = \emptyset$, then $A \times B$ is the set

$$\{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\}$$

and

$$A \times C = \emptyset.$$

We define the ***Cartesian product of*** n ***sets*** to be

$$A_1 \times \cdots \times A_n = \{(a_1, \ldots, a_n) : a_i \in A_i \text{ for } i = 1, \ldots, n\}.$$

If $A = A_1 = A_2 = \cdots = A_n$, we often write A^n for $A \times \cdots \times A$ (where A would be written n times). For example, the set $\mathbb{R}^3$ consists of all of 3-tuples of real numbers.

Subsets of $A \times B$ are called ***relations***. We will define a ***mapping*** or ***function*** $f \subset A \times B$ from a set A to a set B to be the special type of relation where each element $a \in A$ has a unique element $b \in B$ such that $(a, b) \in f$. Another way of saying this is that for every element in A, f assigns a unique element in B. We usually write $f : A \to B$ or $A \xrightarrow{f} B$. Instead of writing down ordered pairs $(a, b) \in A \times B$, we write $f(a) = b$ or $f : a \mapsto b$. The set A is called the ***domain*** of f and

$$f(A) = \{f(a) : a \in A\} \subset B$$

is called the ***range*** or ***image*** of f. We can think of the elements in the function's domain as input values and the elements in the function's range as output values.

where

$$
\begin{aligned}
&F_{11}(u_n, v_n, \delta) \\
&= -\frac{(1+b)x^*\delta}{2a} - \frac{a\sin x^* u_n^2}{2} - \frac{(1+b)x^*\delta^2}{4a^2} \\
&\quad - \frac{a\cos x^* u_n^3}{6} - \frac{(1+b)x^*\delta^3}{8a^3} + O(\|(u_n, v_n, \delta)\|^4), \\
&F_{12}(u_n, v_n, \delta) \\
&= -\frac{\left(-x^*b^2 + 2a\sin x^* - 2x^*b - x^*\right)\delta}{2a} \\
&\quad + \frac{a(b-1)\sin x^* u_n^2}{2} + \frac{(1+b)^2 x^*\delta^2}{4a^2} - \cos x^*\delta u_n \\
&\quad + \frac{a(b-1)\cos x^* u_n^3}{6} + \frac{(1+b)^2 x^*\delta^3}{8a^3} + \frac{\sin x^*\delta u_n^2}{2} \\
&\quad + O(\|(u_n, v_n, \delta)\|^4).
\end{aligned}
$$

Consider the following system

$$
\begin{pmatrix} u_{n+1} \\ v_{n+1} \\ \delta_{n+1} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_n \\ v_n \\ \delta_n \end{pmatrix} + \begin{pmatrix} \widehat{F}_{11}(u_n, v_n, \delta_n) \\ \widehat{F}_{12}(u_n, v_n, \delta_n) \\ 0 \end{pmatrix}, \tag{15}
$$

where

$$
a_{11} = -1 + a\cos x^*, \quad a_{12} = 1,
$$

then the system (15) becomes

$$
\begin{aligned}
\begin{pmatrix} \tilde{x}_{n+1} \\ \tilde{y}_{n+1} \\ \tilde{\delta}_{n+1} \end{pmatrix} &= \begin{pmatrix} -1 & 0 & 0 \\ 0 & -b + a\cos x^* & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{x}_n \\ \tilde{y}_n \\ \tilde{\delta}_n \end{pmatrix} \\
&\quad + \begin{pmatrix} \tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ \tilde{F}_{12}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ 0 \end{pmatrix},
\end{aligned} \tag{16}
$$

where

$$
\begin{aligned}
&\tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\
&= -\frac{\tan x^* \sec x^* \tilde{x}_n^2}{2a} - \frac{\tan x^* \tilde{x}_n \tilde{y}_n}{b-1} + \frac{\sin x^* \tan x^* \tilde{\delta}_n \tilde{x}_n}{2\cos x^*(a-2b-2)} \\
&\quad - \frac{a\sin x^* \tilde{y}_n^2}{2(b-1)^2} + \frac{a\sin^2 x^* \tilde{y}_n \tilde{\delta}_n}{2(b-1)(a\cos x^* - b - 1)} \\
&\quad - \frac{\tilde{\delta}_n^2}{8(a\cos x^* - b - 1)^2 a^2}\Big(\big(2x^*(b+1)^3 + a^3\sin^3 x^* \\
&\quad - 4ax^*(b+1)^2\cos x^* + 2a^2x^*(b+1)\cos^2 x^*\big)\Big) \\
&\quad + \frac{\sec^2 x^* \tilde{x}_n^3}{6a^2} + \frac{\sec x^* \tilde{x}_n^2 \tilde{y}_n}{2a(b-1)} - \frac{\tan x^* \tilde{x}_n^2 \tilde{\delta}_n}{4a(a\cos x^* - b - 1)}
\end{aligned}
$$

[redacted]

Lecture Note (p.1)

第十一届清疏大学生数学竞赛班讲义

非数学类 & 数学类共用

作者：清疏数学

组织：清疏大学

时间：July 7, 2024

版本：2.0

作者联系方式：QQ:937821418

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Limits, Series, and Fractional Part Integrals

Problems in Mathematical Analysis

Springer

block_layout_regression_v4

8.34

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MATHCRAFT_FORCE_ORT_CPU MATHCRAFT_HOME

(永久生效):